MIL-S-19500/162B
27 July 1964
SUPERSEDING
MIL-S-19500/162A(EL)
25 January 1963

MILITARY SPECIFICATION

SEMICONDUCTOR DEVICES, DIODE, SILICON, POWER RECTIFIER

JAN TYPES 1N1614, 1N1615, 1N1616, 1N4458, 1N4459, 1N1614R, 1N1615R, 1N1616R, 1N4458R, 1N4459R

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for silicon, power rectifiers with the following ratings.

1.2 Reverse types (suffix R). Reverse and normal types are identical except: The normal types have the cathode connected to the stud; the reverse types have the anode connected to the stud. Designated values are applicable to both types.

1.3 Maximum ratings.

Type	V_r	V_{RM}(wkg)	i_f	I_o 1/ T_C = 25°C	I_o (T_C = 150°C)	i_f(surge) (at: 1/120 sec)
	v	v(pk)	a	Adc	Adc	a
1N1614	240	200	15	10	5	150
1N1615	480	400	15	10	5	150
1N1616	720	600	15	10	5	150
1N4458	960	800	15	10	5	150
1N4459	1200	1000	15	10	5	150

1/ Derate at 40 mAdc/°C above T_C = 25°C

OPERATING CASE TEMPERATURE: -65°C to +150°C

STORAGE TEMPERATURE: -65°C to +200°C

BAROMETRIC PRESSURE, REDUCED: 8 mm Hg at $V_{RM} \leq 600$ Vdc; 15 mm Hg at $V_{RM} > 600$ Vdc

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARDS

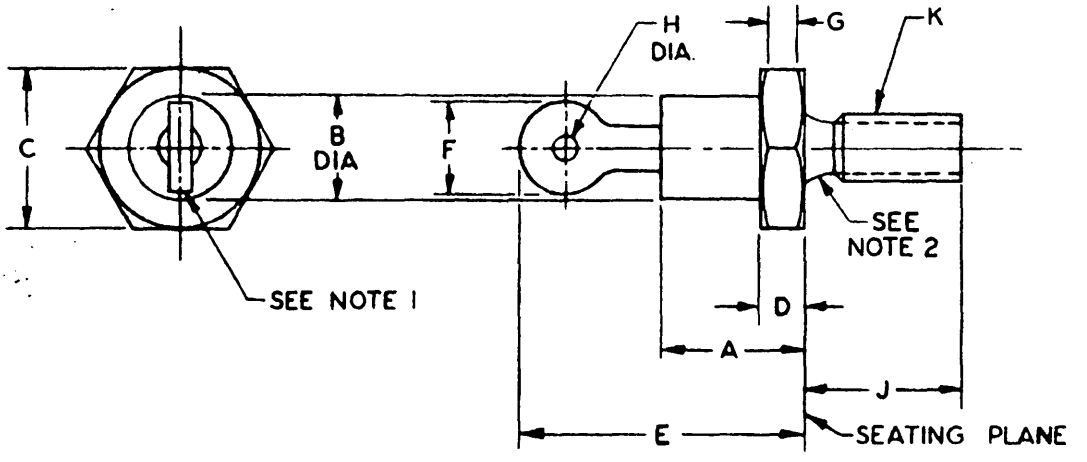
Military

MIL-STD-750 - Military Standard Test Methods for Semiconductor Devices.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 5960

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DIMENSIONS					NOTES
LTR	INCHES		MILLIMETERS		
	MIN	MAX	MIN	MAX	
A		.405		10.29	
B		.424		10.77	
C	.424	.437	10.77	11.10	
D	.075	.175	1.90	4.44	
E		.800		20.32	
F		.250		6.35	7
G	.060		1.52		
H	.060		1.52		
J	.422	.453	10.72	11.51	
K					4,5,6

NOTES:
1. Angular orientation of this terminal is undefined.
2. Dia of unthreaded portion .189 inch (4.80 mm) max; .163 inch (4.14 mm) min.
3. The A.S.A. thread reference is 10–32 UNF2A.
4. Max pitch dia of plated threads shall be basic pitch dia .169 inch (4.31 mm).
5. Unit shall not be damaged by torque of 15 in-lb applied to 10–32NF2B nut assembled on thread.
6. Complete threads shall extend to within 2-1/2 threads of the seating plane.
7. Terminal end shape is unrestricted.
8. Reversed (anode to stud) units shall be marked with an R following the last digit in the type no
9. For marking see 3.5.
10. Metric equivalents are shown for general information only and are based upon 1 inch = 25.4 mm to the nearest .01 millimeter.

Figure 1. Semiconductor devices, diode, types 1N1614, 1N1615, 1N1616, 1N4458, 1N4459, and the reverse types (suffix R added).

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3. REQUIREMENTS

3.1 General. Requirements shall be in accordance with Specification MIL-S-19500, and as specified herein.

3.2 Abbreviations and symbols. The abbreviations and symbols used herein are defined in Specification MIL-S-19500.

3.3 Design and construction. The semiconductor diodes shall be of the design, construction, and physical dimensions specified in figure 1.

3.4 Performance characteristics. Performance characteristics shall be as specified in tables I, II, and III.

3.5 Marking. The marking shall be placed on each device in accordance with MIL-S-19500. The polarity shall be indicated by graphic symbol with the arrow pointing toward the negative end for forward bias.

3.5.1 The manufacturer, at his option, may omit the following marking:

Country of origin.

3.6 Salt spray (corrosion). The diode shall be examined for evidence of corrosion and legibility of marking before the specified measurements are made (see 4.4.3).

3.7 Solderability. The immersion depth shall cover the flat portion of the terminals.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with Specification MIL-S-19500, and as specified herein.

4.2 Qualification inspection. Qualification inspection shall consist of the examination and tests specified in tables I, II, and III. Qualification testing of either polarity is sufficient to obtain qualification approval of both polarities.

4.3 Quality conformance inspection. Quality conformance inspection shall consist of the examinations and tests specified in groups A, B and C. Group A inspection shall be performed on a sublot basis. Group B inspection shall be performed on a lot basis per MIL-S-19500. Subgroups 7 and 8 of group B and group C inspections shall be performed on a sublot basis or on the highest voltage type present in the lot. In the event subsequent lots contain higher voltage devices than have previously been accepted under group C provisions, these higher voltage devices shall be subjected to group C inspection. A failure of these higher voltage devices does not invalidate previously established group C status on lower voltage devices.

4.3.1 Group A inspection. Group A inspection shall consist of the examinations and tests specified in table I.

4.3.2 Group B inspection. Group B inspection shall consist of the examinations and tests specified in table II.

4.3.3 Group C inspection. Group C inspection shall consist of the examinations and tests specified in in table III.

4.4 Methods of examination and test. Methods of examination and test shall be as specified in tables I, II, and III, and as follows:

4.4.1 Pulse measurement. Pulse measurements shall be in accordance with section 4 of MIL-STD-750.

4.4.2 Steady-state operation life. This test shall be conducted with a half-sine wave of the specified peak voltage impressed across the diode in the reverse direction followed by a half-sine wave form of the specified average rectified current. The forward conduction angle of the rectified current shall not be greater than 180° nor less than 130°; and the power shall be equal to or greater than that of a half-sine wave.

4.4.3 Salt spray (corrosion). The markings shall be legible after the test. There shall be no evidence (when examined with no magnification) of flaking or pitting of the finish or corrosion that will interfere with the mechanical and electrical application of the devices.

4.4.4 Inspection conditions. Unless otherwise specified herein, all inspections shall be made at case temperature (T_C) of 25° ±3°C.

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Table I. Group A inspection.

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		
	Method	Details			Min	Max	Unit
Subgroup 1			7				
Visual and mechanical examination	2071			---	---	---	---
Subgroup 2			5				
Forward voltage (See 4.4.1)	4011	I_F = 15 Adc		V_F	---	1.5	Vdc
Reverse current	4016			I_R			
1N1614		V_R = 200 Vdc			---	50	uAdc
1N1615		V_R = 400 Vdc			---	50	uAdc
1N1616		V_R = 600 Vdc			---	50	uAdc
1N4458		V_R = 800 Vdc			---	50	uAdc
1N4459		V_R = 1000 Vdc			---	50	uAdc
Reverse current	4016	T_C = 150°C		I_R			
1N1614		V_R = 200 Vdc			---	500	uAdc
1N1615		V_R = 400 Vdc			---	500	uAdc
1N1616		V_R = 600 Vdc			---	500	uAdc
1N4458		V_R = 800 Vdc			---	500	uAdc
1N4459		V_R = 1000 Vdc			---	500	uAdc
Reverse current at peak reverse voltage:	4016			i_r			
1N1614		v_r = 240 v(pk)			---	500	ua(pk)
1N1615		v_r = 480 v(pk)			---	500	ua(pk)
1N1616		v_r = 720 v(pk)			---	500	ua(pk)
1N4458		v_r = 960 v(pk)			---	500	ua(pk)
1N4459		v_r = 1200 v(pk)			---	500	ua(pk)

Table II. Group B inspection.

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		
	Method	Details			Min	Max	Unit
Subgroup 1			10				
Physical dimensions	2066			---	---	---	---
Subgroup 2			10				
Solderability (see 3.7)	2026	Dwell time = 10 ± 1 sec					
Temperature cycling	1051	Test cond. F		---	---	---	---
Thermal shock (glass strain)	1056	Test cond. A		---	---	---	---
Terminal strength: Tension	2036	Test cond. A; 2 lbs; t = 15 ± 3 sec		---	---	---	---
Torque (seal)		Test cond. D_1; 10 oz. in.; t = 15 ± 3 sec		---	---	---	---
Bending stress (axial strain)		Test cond. F; method B, 3 lbs; t = 15 ± 3 sec		---	---	---	---
Moisture resistance	1021	Omit initial conditioning		---	---	---	---
End points: Forward voltage (see 4.4.1)	4011	I_F = 15 Adc		V_F	---	1.5	Vdc

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Table II. Group B inspection. - Continued

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		
	Method	Details			Min	Max	Unit
Subgroup 2 (Cont)							
End points: (Cont)							
Reverse current:	4016			I_R			
1N1614		V_R = 200 Vdc			---	50	uAdc
1N1615		V_R = 400 Vdc			---	50	uAdc
1N1616		V_R = 600 Vdc			---	50	uAdc
1N4458		V_R = 800 Vdc			---	50	uAdc
1N4459		V_R = 1000 Vdc			---	50	uAdc
Subgroup 3			10				
Shock	2016	500 G, t ≅ 1 msec; 5 blows each orientation: X_1, Y_1, and Y_2		---	---	---	---
Vibration fatigue (nonoperating)	2046	10 G		---	---	---	---
Vibration, variable frequency (nonoperating)	2056	10 G		---	---	---	---
Constant acceleration	2006	10,000 G, X_1, Y_1, and Y_2 orientations		---	---	---	---
End points:							
(Same as subgroup 2)							
Subgroup 4			10				
Terminal strength:	2036						
Torque (stud)		Test cond. D_2; 15 lbs. in. t = 15 ± 3 sec		---	---	---	---
End points:							
(Same as subgroup 2)							
Subgroup 5			10				
Surge current	4066	i_f(surge) = 150 a; I_o = 5 Adc; T_C = 150°C; 1 surge/minute, t_p = 1/120 sec, 5 cycles					
1N1614		V_{RM}(wkg) = 200 v(pk)		---	---	---	---
1N1615		V_{RM}(wkg) = 400 v(pk)		---	---	---	---
1N1616		V_{RM}(wkg) = 600 v(pk)		---	---	---	---
1N4458		V_{RM}(wkg) = 800 v(pk)		---	---	---	---
1N4459		V_{RM}(wkg) = 1000 v(pk)		---	---	---	---
End points:							
(Same as subgroup 2)							
Subgroup 6			10				
Salt spray (corrosion) (see 3.6)	1046	Test cond. B		---	---	---	---
End points:							
(Same as subgroup 2)							

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Table II. Group B inspection. - Continued

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		
	Method	Details			Min	Max	Unit
Subgroup 7			λ = 10				
High-temperature life (non-operating)	1031	T_A = 200°C		---	---	---	---
End points:							
Forward voltage (see 4.4.1)	4011	I_F = 15 Adc		V_F	---	1.5	Vdc
Reverse current:	4016			I_R			
1N1614		V_R = 200 Vdc			---	100	uAdc
1N1615		V_R = 400 Vdc			---	100	uAdc
1N1616		V_R = 600 Vdc			---	100	uAdc
1N4458		V_R = 800 Vdc			---	100	uAdc
1N4459		V_R = 1000 Vdc			---	100	uAdc
Subgroup 8			λ = 10				
Steady state operation: (see 4.4.2)	1026	T_C = 150°C; I_o = 5 Adc, f = 60 cps					
1N1614		V_{RM}(wkg) = 200 v(pk)			---	---	---
1N1615		V_{RM}(wkg) = 400 v(pk)			---	---	---
1N1616		V_{RM}(wkg) = 600 v(pk)			---	---	---
1N4458		V_{RM}(wkg) = 800 v(pk)			---	---	---
1N4459		V_{RM}(wkg) = 1000 v(pk)			---	---	---
End points:							
(Same as subgroup 7)							
Subgroup 9			10				
Low-temperature operation:		T_A = -65°C					
Forward voltage (see 4.4.1)	4011	I_F = 15 Adc		V_F	---	1.7	Vdc
Reverse current:	4016			I_R			
1N1614		V_R = 200 Vdc			---	100	uAdc
1N1615		V_R = 400 Vdc			---	100	uAdc
1N1616		V_R = 600 Vdc			---	100	uAdc
1N4458		V_R = 800 Vdc			---	100	uAdc
1N4459		V_R = 1000 Vdc			---	100	uAdc

Table III. Group C inspection.

Examination or test	MIL-STD-750		LTPD	Symbols	Limits		
	Method	Details			Min	Max	Unit
Subgroup 1 1/			10				
Barometric pressure, reduced (altitude operation)	1001	8 mm Hg at $V_{RM} \leq$ 600 Vdc; 15 mm Hg at $V_{RM} >$ 600 Vdc; t = 60 sec		---	---	---	---
Measurement during test:							
Reverse current:	4016			I_R			
1N1614		V_R = 200 Vdc			---	50	uAdc
1N1615		V_R = 400 Vdc			---	50	uAdc
1N1616		V_R = 600 Vdc			---	50	uAdc
1N4458		V_R = 800 Vdc			---	50	uAdc
1N4459		V_R = 1000 Vdc			---	50	uAdc

1/ This test shall be conducted on the initial lot and thereafter every six months.

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5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall conform to Specification MIL-S-19500.

6. NOTES

6.1 The notes specified in Specification MIL-S-19500 are applicable to this specification.

Custodians:
Army - EL
Navy - Ships
Air Force -11

Preparing activity:
Army - EL

(Project 5960-1833)

Reviewers:
Army - EL
Navy - SH
Air Force - 11, 17, 85

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